

TAI, Wing-Kuen (William)

+852 5928 8549 | williamtaiwk@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | Shatin / Ma On Shan, Hong Kong

SUMMARY

Computer Engineering graduate from HKUST specializing in Backend Systems and AI Integration. Experienced in rapid end-to-end prototyping, RESTful API design (Python FastAPI), and LLM agent toolchains (Model Context Protocol, Function Calling). Skilled in deploying containerized microservices with Docker and building reliable applications across web and mobile platforms.

EDUCATION

The Hong Kong University of Science and Technology (HKUST)

Sep 2023 – Jun 2026

Bachelor of Engineering in Computer Engineering

• **Practical University Training:** Linux System & Network Administration, Cloud Services in Azure (HKUST Industrial Training)

HKU SPACE Community College

Sep 2021 – Aug 2023

Associate of Engineering in Computer Science

TECHNICAL SKILLS

Programming Languages: Python, Kotlin, TypeScript, JavaScript, SQL, C++, Java, Dart

AI & Agent Orchestration: LLM Function Calling / Tool Interception, Model Context Protocol (MCP), Hugging Face APIs, Prompt & Schema Engineering, Structured Output Parsing

Backend & Integration: Python (FastAPI), RESTful API Design, Docker, Docker Compose, Linux (Ubuntu), Microservice Integration, Azure (SQL, Key Vault)

Databases & Storage: MongoDB Atlas, GridFS (Binary Storage), MySQL, SQLite / Room

Web & Mobile: Next.js, React, TypeScript, Kotlin (Jetpack Compose), Flutter, Tailwind CSS

Developer Tools: Git, GitHub Actions (CI/CD Pipelines), Vitest, Postman

TECHNICAL PROJECTS

AI-Powered Multi-Modal Health Platform

Jun 2025 – May 2026

HKUST Final Year Project (Grade A) | Backend Lead & Systems Developer (4-person team)

- **Engineered modular FastAPI backend** on Ubuntu serving 30+ RESTful endpoints for user auth, OCR nutrition analytics, gamified systems, and social leaderboards
- **Designed and integrated multi-modal LLM pipeline** with Qwen-2.5-VL-72B via Hugging Face Router, implementing zero-shot schema constraints and regex sanitization to ensure deterministic JSON parsing
- **Built binary media storage service** using MongoDB GridFS, handling image uploads and asynchronous retrieval pipelines
- **Containerized deployment architecture** using Docker Compose with volume-mounted dependencies for OpenCV/OCR, ensuring reproducible cross-environment setup

Smart Trip Planner (Agentic Travel App)

Sep – Nov 2025

Mobile & Cloud Integration | Flutter, Gemini Tool Calling, MongoDB Atlas

- **Implemented LLM tool-calling loop** using Gemini 2.5 Flash Lite to dynamically intercept user queries, trigger external flight/hotel APIs (Amadeus), and resolve airport IATA codes
- **Built automated data feedback flow** by returning API payloads as structured FunctionResponses, synthesizing multi-day itineraries with Google Places integrations
- **Structured state management** using MVVM architecture with Provider for deterministic UI updates and user session persistence in MongoDB Atlas

AI Developer Toolchain & MCP Integration

Jan – Feb 2026

Personal Project | Model Context Protocol (MCP), Docker, Python, Git

- **Deployed containerized Model Context Protocol (MCP) servers** using Docker to securely bridge local AI coding assistants with GitHub APIs and local filesystem operations
- **Engineered structured JSON schema tool interfaces**, enabling autonomous AI agents to execute multi-step repository inspections, file refactoring, and automated Git workflows safely

TimeLock Anti-Addiction Android Application

Aug 2026

Personal Project | Kotlin, Jetpack Compose, Room / SQLite, Material 3

- **Built a state-machine and persistent countdown** using StateFlow and a Room heartbeat, preventing background coroutine freezes and timing drift across app switches
- **Implemented global cooldown and 60-second emergency pass** to curb cross-app usage, enforcing single-use tokens to prevent cooldown bypass
- **Resolved Android edge cases and UI consistency**, including Xiaomi background event suppression, WindowMetrics orientation clamping, and Material 3 theme injection for WindowManager overlays

React Testing & CI Pipeline Showcase

Sep 2026

Personal Project | Vitest, React Testing Library, MSW, GitHub Actions

- **Wrote integration test suites** using Vitest and React Testing Library for authentication pages, using Mock Service Worker (MSW) to simulate 401, 404, and 500 network error states
- **Configured GitHub Actions CI workflow** with npm caching to run automated test suites and collect coverage artifacts on pull requests

Interactive Portfolio Web Application

Jun 2026

Personal Project | Next.js, React, TypeScript, Tailwind CSS

- **Built responsive portfolio** with Next.js App Router, implementing dynamic video lazy-loading to optimize First Contentful Paint (FCP)

WORK EXPERIENCE

IT Deployment Technician (Contract) | Proseek Technology

Sep 2026 – Present

Enterprise Infrastructure Project (Client: UBS, West Kowloon HQ)

- **Deployed enterprise workstations** and dual-monitor displays across UBS trading floors, setting up hardware, power distribution, and cable management to bank workspace standards
- **Audited hardware assets** by checking BIOS configurations and recording serial numbers as part of a 500+ endpoint enterprise rollout

Office Assistant | Hayes Cleveland Limited

Jul 2024 – Nov 2024

- **Managed inventory reconciliation** and order fulfillment tracking via Google Sheets for an active e-commerce storefront
- **Coordinated parcel logistics** and responded to customer order inquiries, maintaining a sub-24h turnaround SLA

Store Assistant | Circle K

Jun 2023 – Aug 2023

- **Executed POS transactions**, cash reconciliations, and inventory restocking in a high-volume retail branch
 - **Handled customer service** and e-commerce pickup fulfillment workflows
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CERTIFICATIONS

• Software Engineering: Modeling Software Systems using UML HKUST via Coursera	Jul 2026
• Gemini Certified University Student Google for Education	Jun 2026
• AI Code Generation for Everyone HKUST via Coursera	Jun 2026
• Search Engines for Web and Enterprise Data HKUST via Coursera	May 2026

LANGUAGES

Cantonese (Native) | English (Fluent) | Mandarin (Fluent)